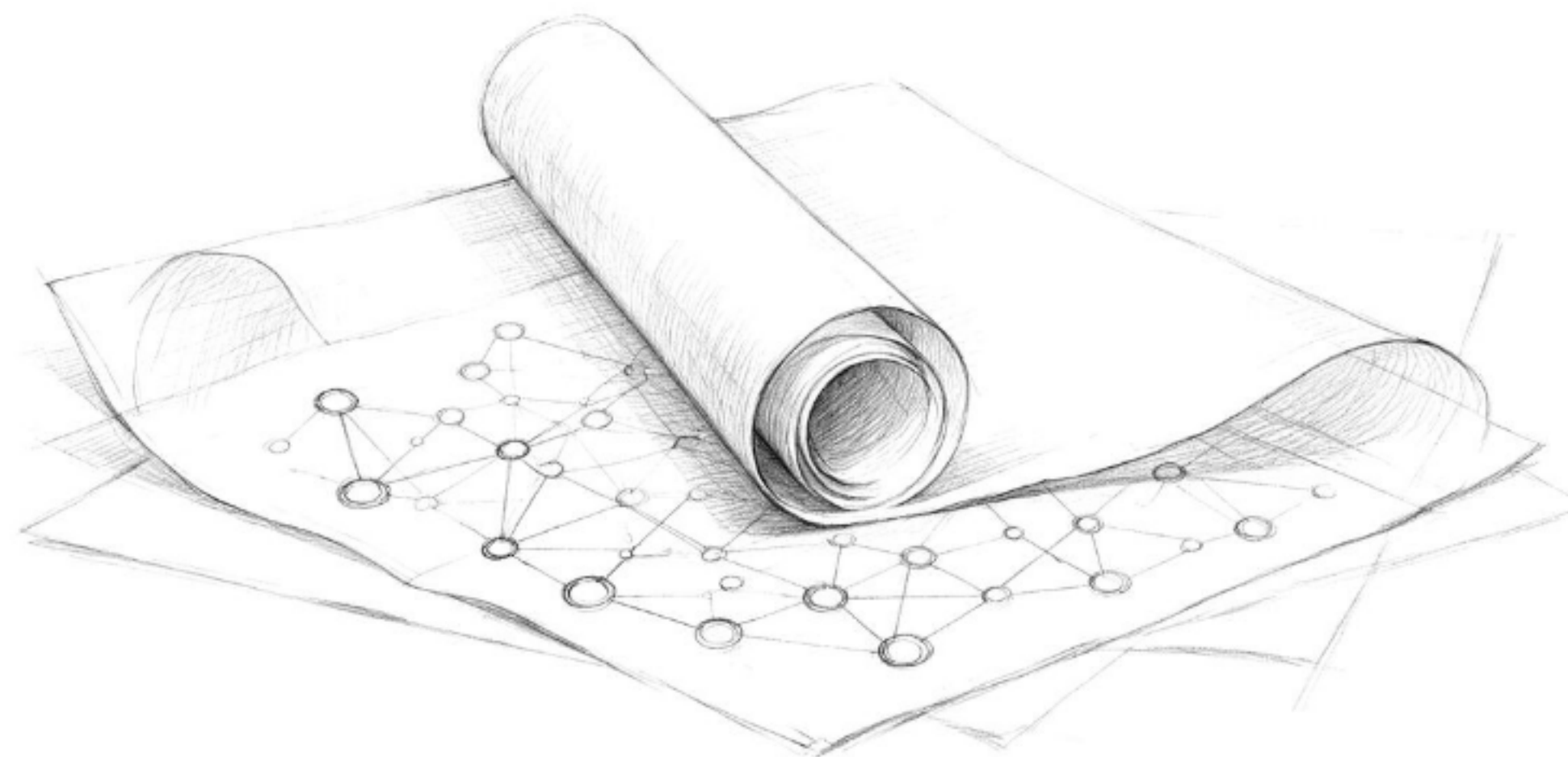
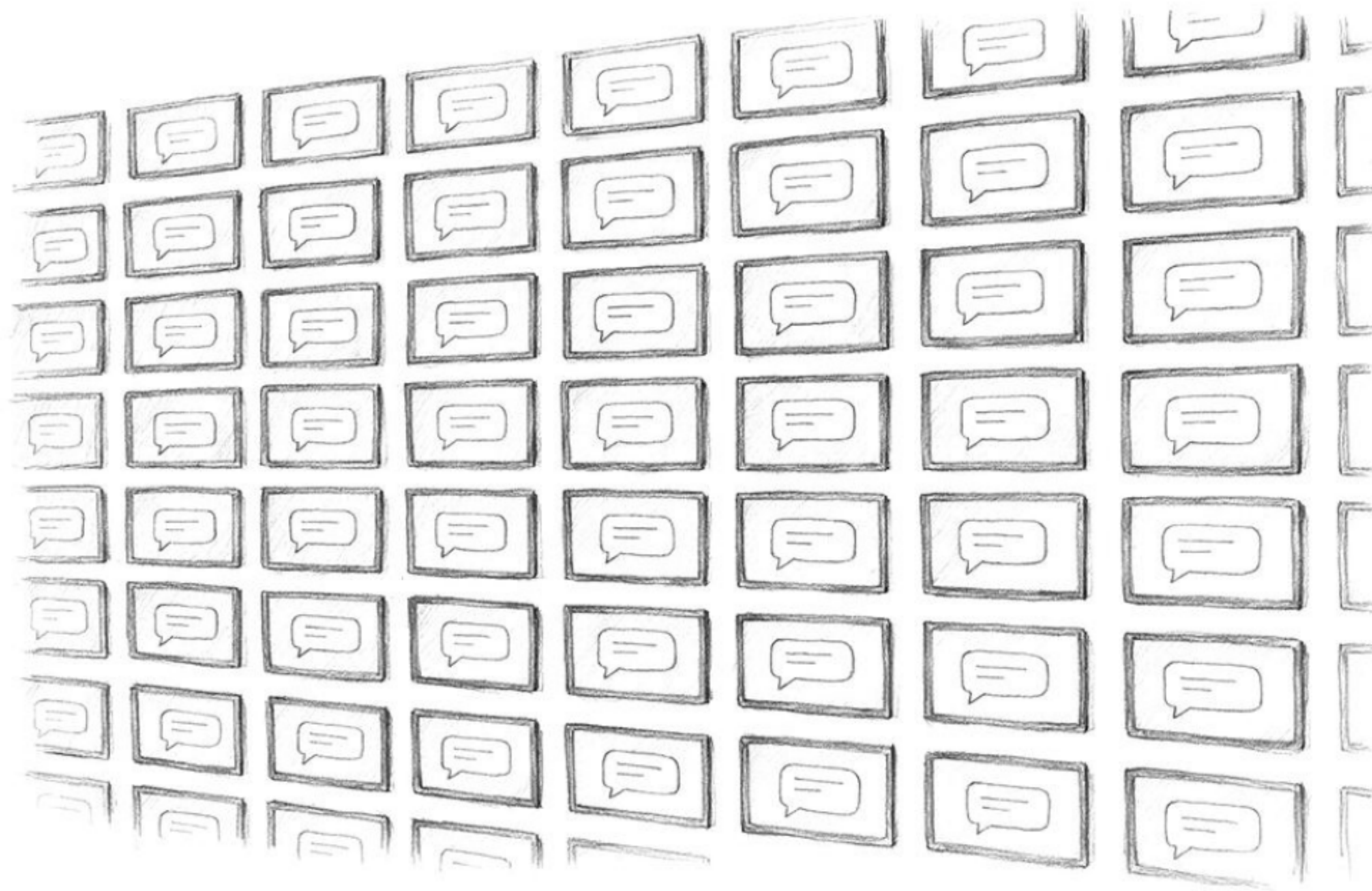


The LLM Revolution — Tokens, Transformers, RAG & Reasoning



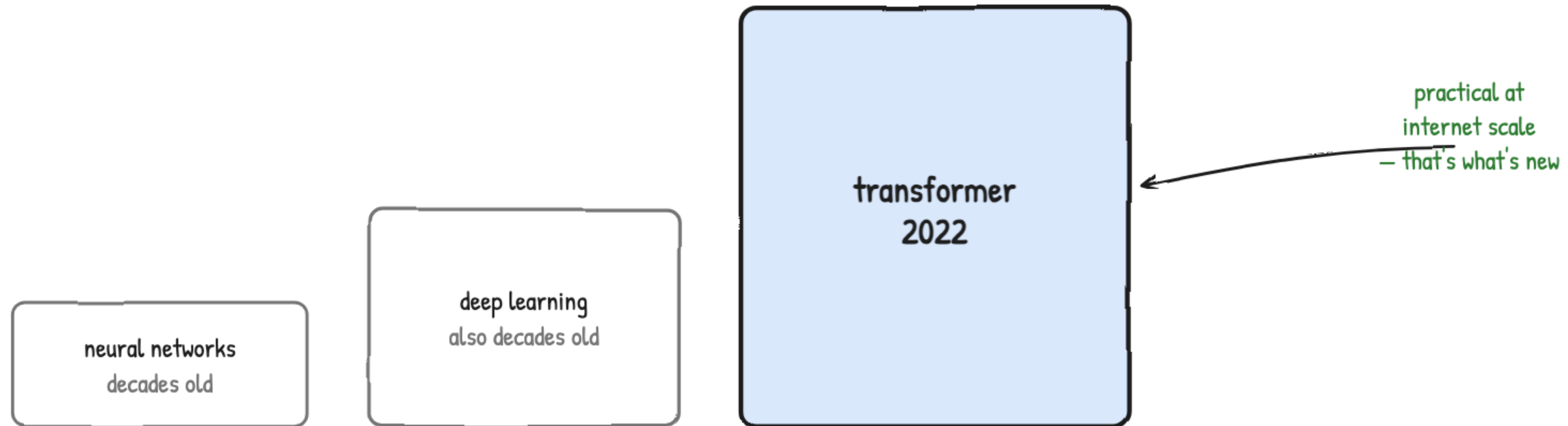
You know half the answer already — this is the other half

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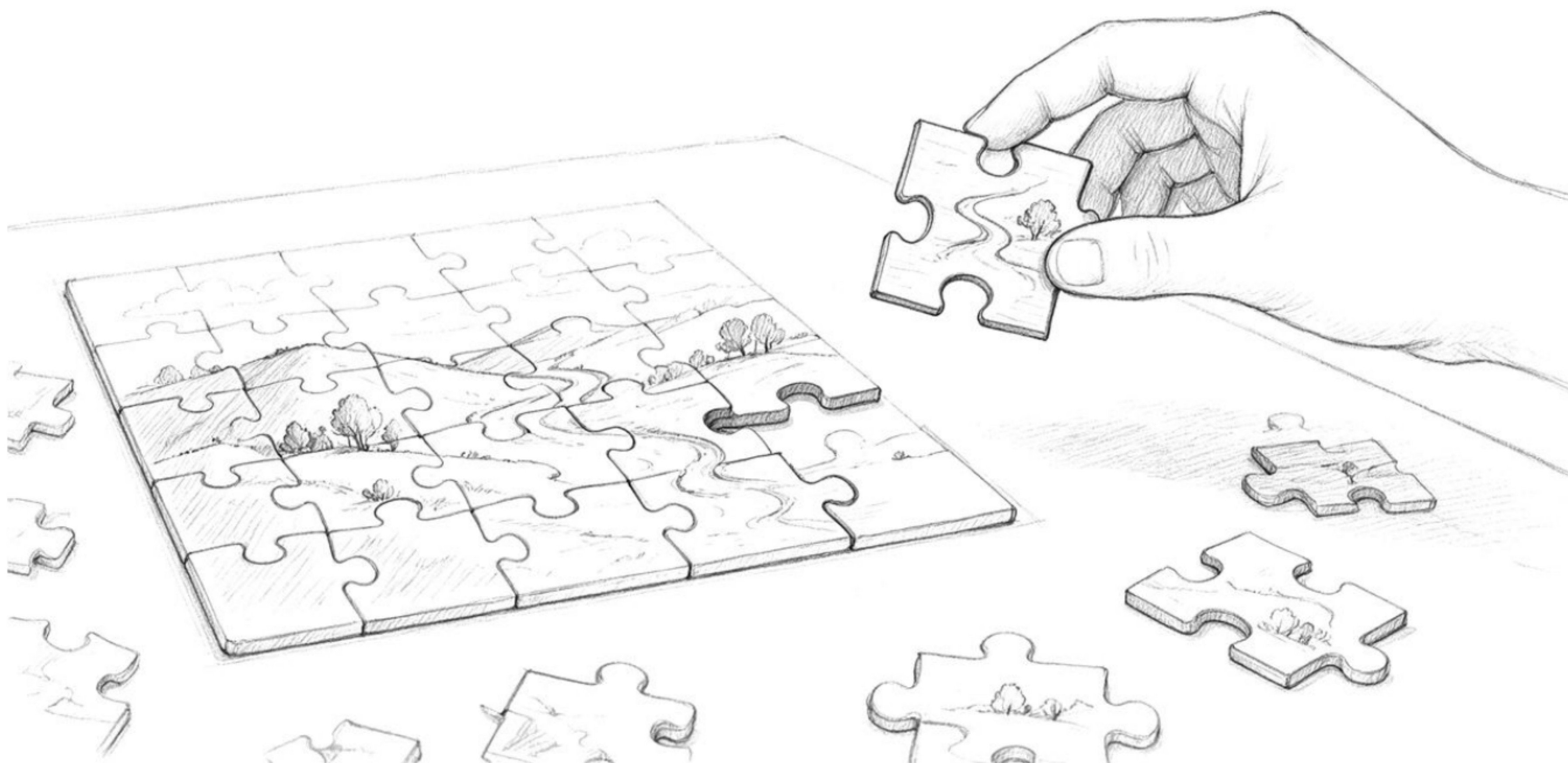


Why did 2022 change everything?

ChatGPT wasn't new math — it was old math at a new scale



Decades-old ideas, made practical at internet scale



Tokens and parameters, one level deeper

A token is rarely a whole word

"disconnecting"



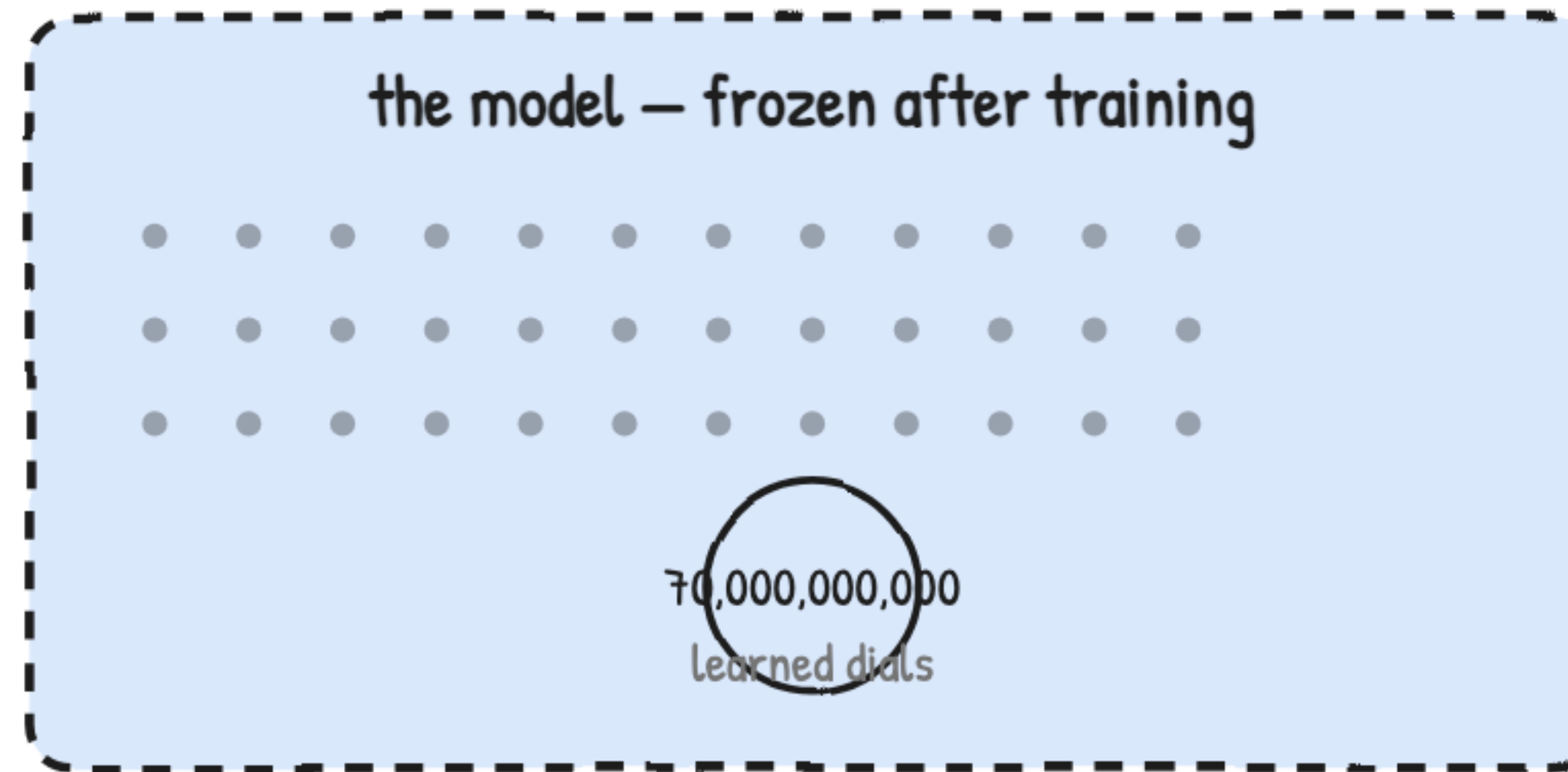
a fixed whole-word list misses every new word — small pieces rebuild almost anything

three or four tokens, one word

billed three or four times, eats the context window

Small reusable pieces rebuild almost any word

Parameters are the dials, and the bill



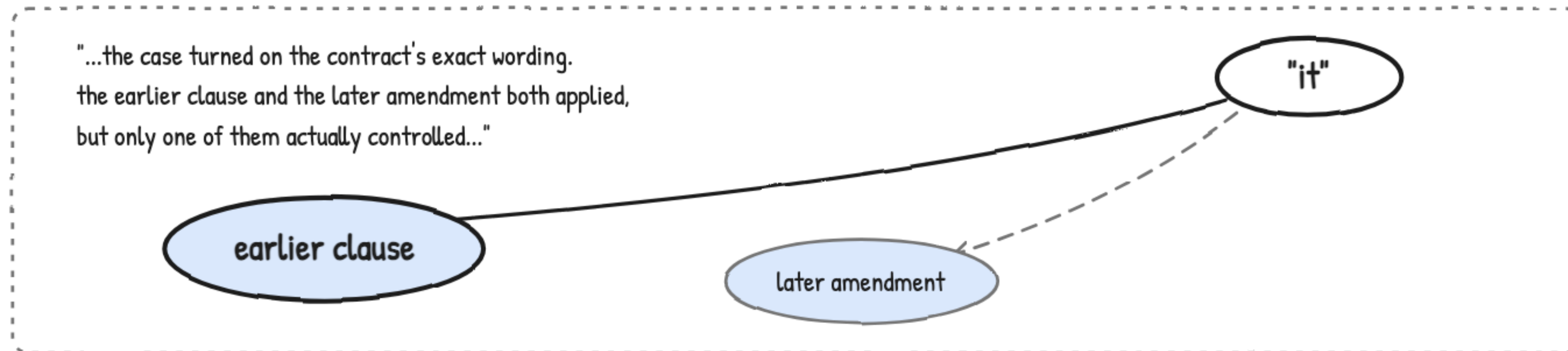
more capacity to represent patterns
and a bigger compute bill — training or running it

More capacity to learn — and a bigger compute bill



Transformers and attention: how a model reads a whole sentence at once

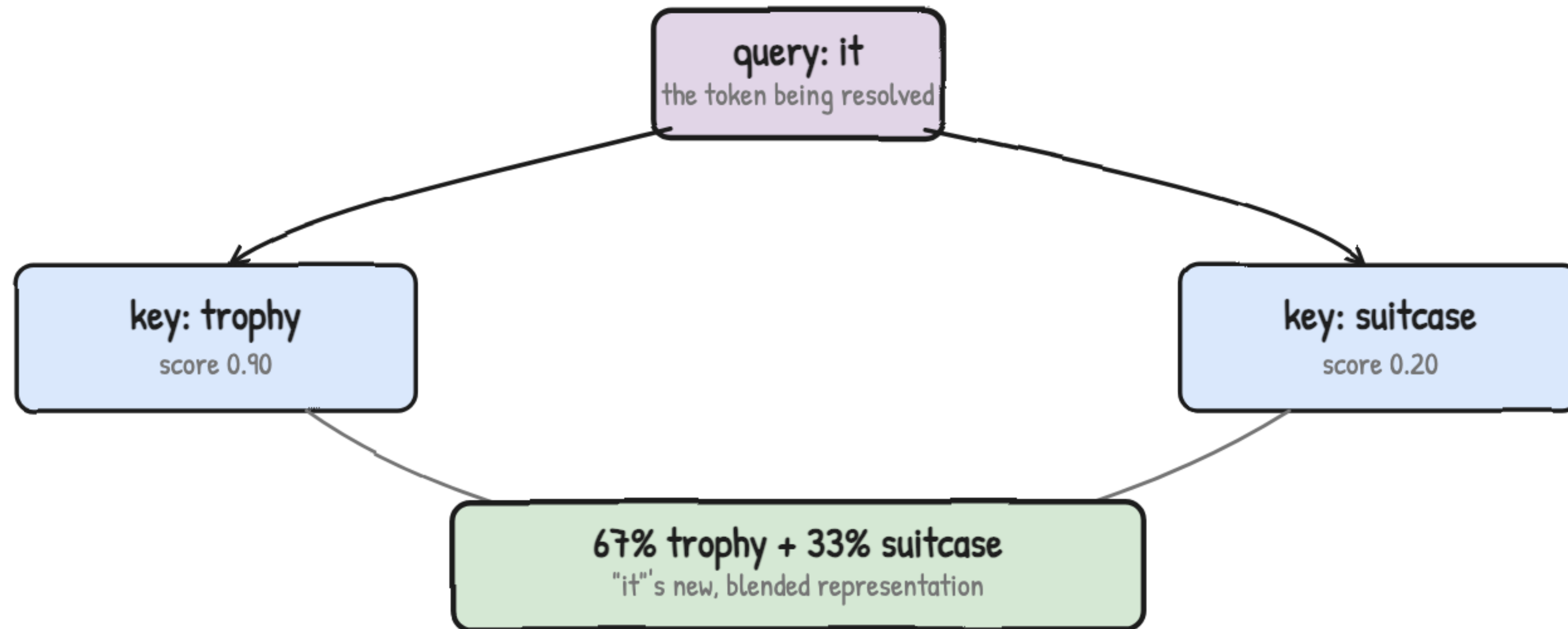
You don't read "it" alone – you glance back and weigh



weighing which one it actually means

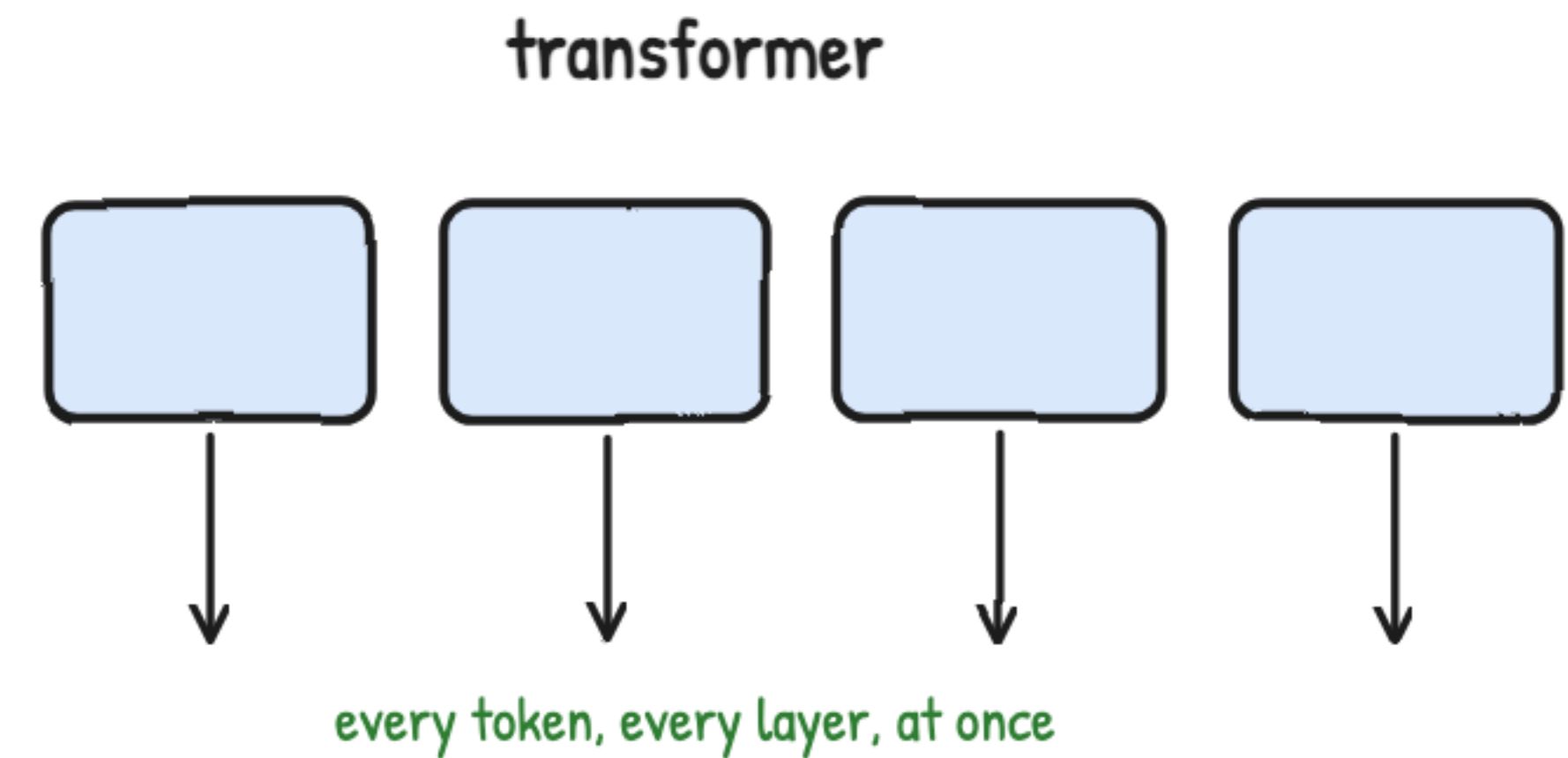
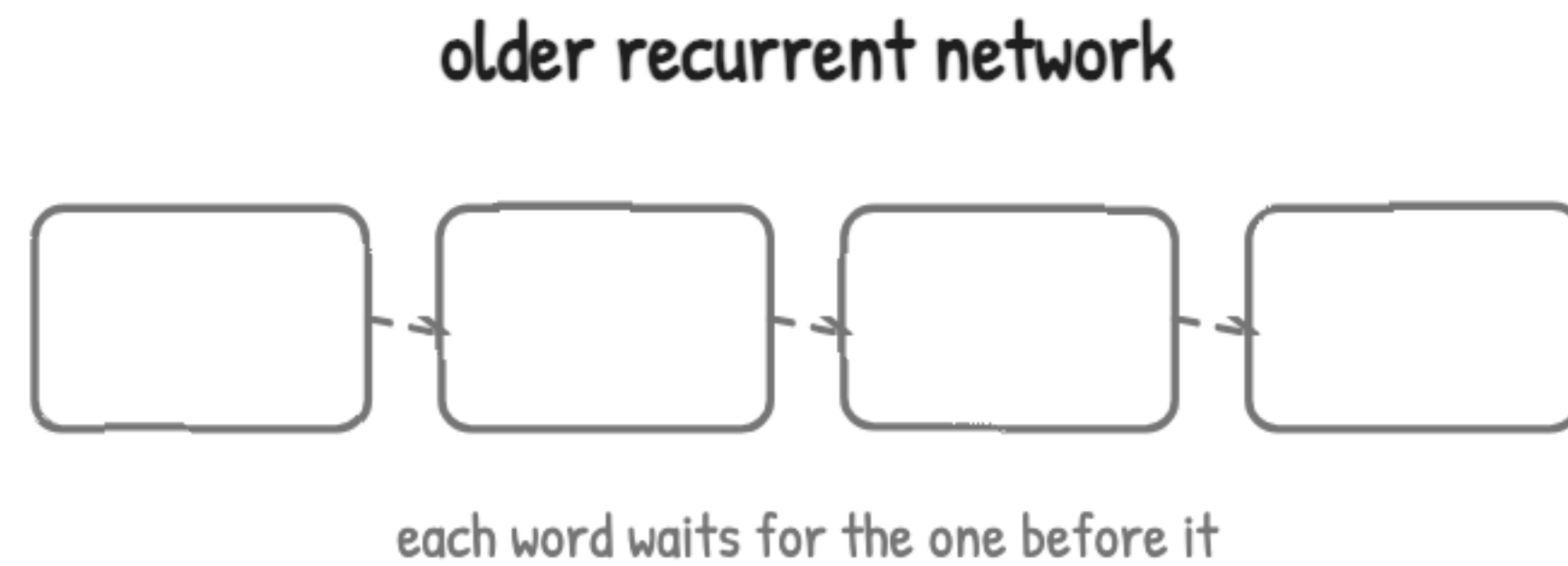
Weighing which earlier phrase it actually means

Attention scores every earlier word, then blends them



Real numbers, not just an analogy

A transformer reads every token at once — no waiting in line



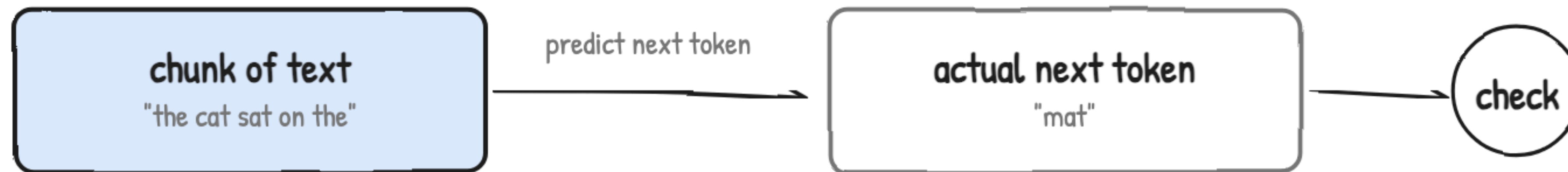
Every token, every layer, at the same time



**Pre-training, fine-tuning, and RLHF – Module 3's
ideas, at internet scale**

Pre-training: predict the next word, then check what came next

supervised learning (Module 3), at internet scale



every sentence on the internet supplies its own label

Every sentence on the internet supplies its own label

Fine-tuning: the same process, a small curated set, one exact behavior

supervised learning again — smaller, curated



same mechanism as pre-training, aimed at one behavior instead of the whole internet

A format, an instruction-following style, a refusal rule

RLHF: human rankings train a reward model, which nudges the LLM

human raters

rank sample answers

this pass turns a raw predictor into something that follows instructions

Module 3's third idea, aimed at the model's own answers

RLHF: human rankings train a reward model, which nudges the LLM



this pass turns a raw predictor into something that follows instructions

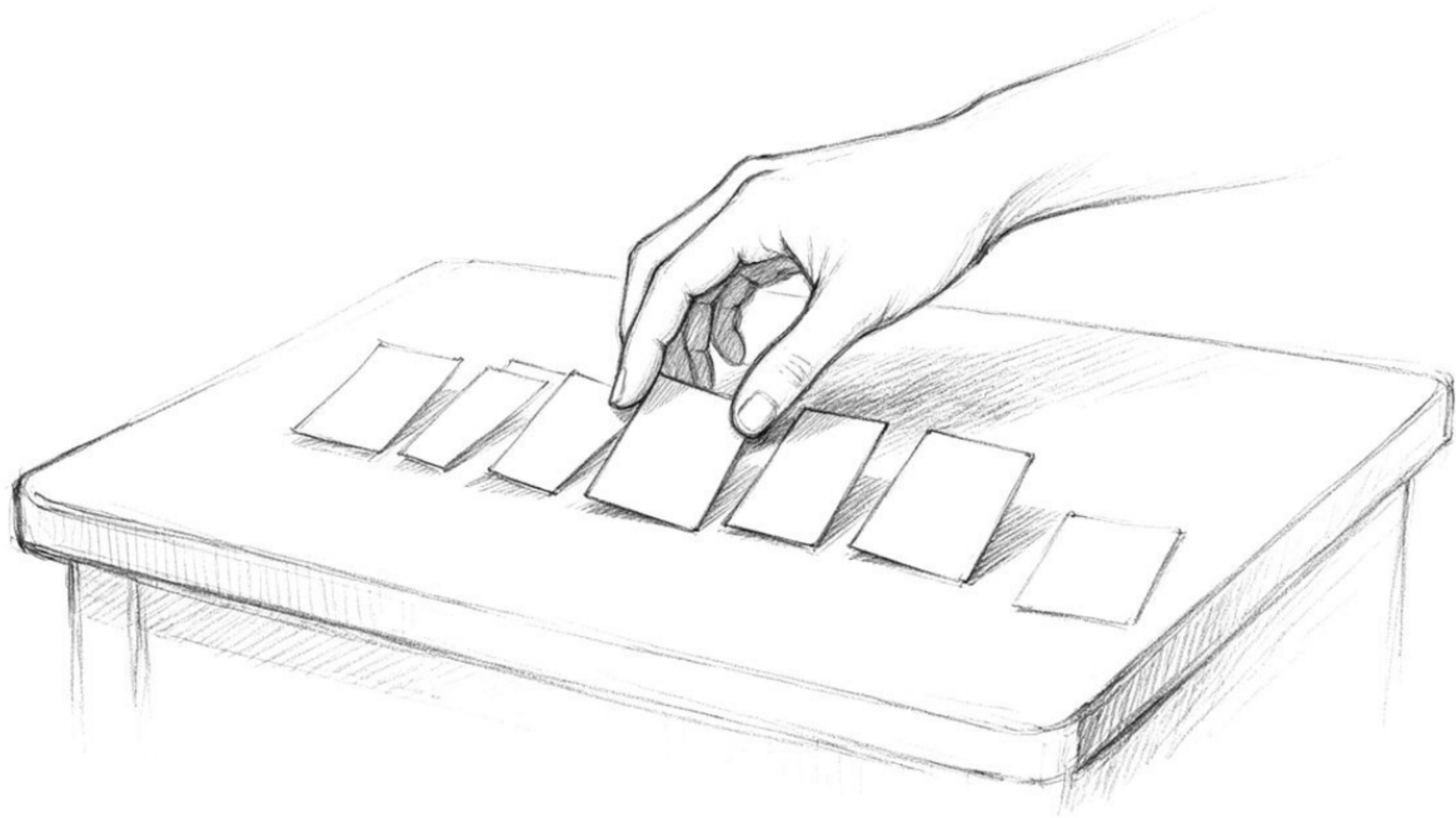
Module 3's third idea, aimed at the model's own answers

RLHF: human rankings train a reward model, which nudges the LLM



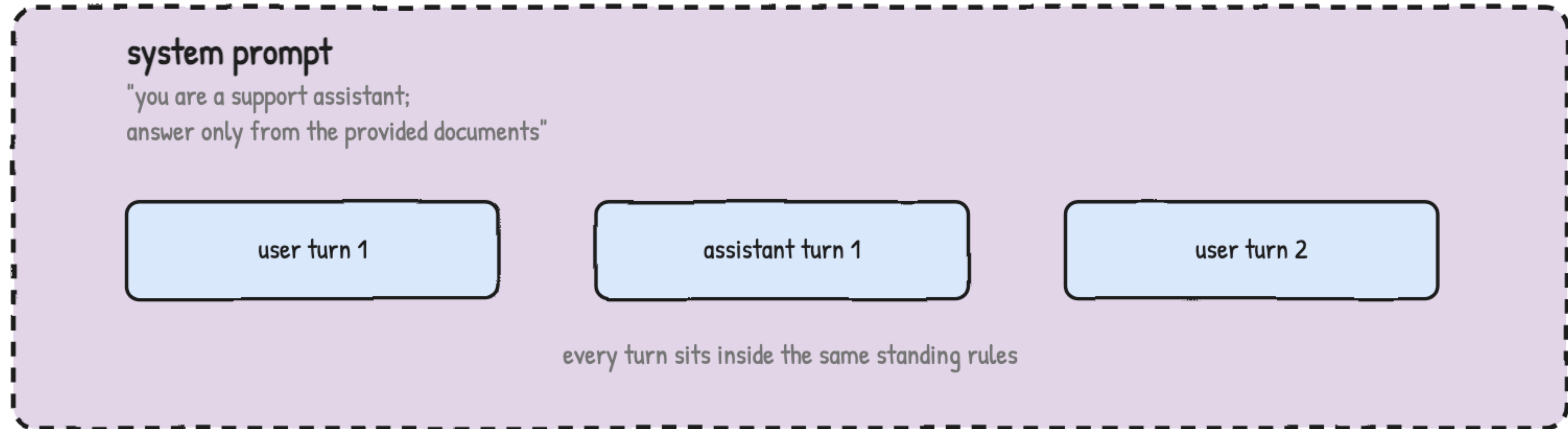
this pass turns a raw predictor into something that follows instructions

Module 3's third idea, aimed at the model's own answers



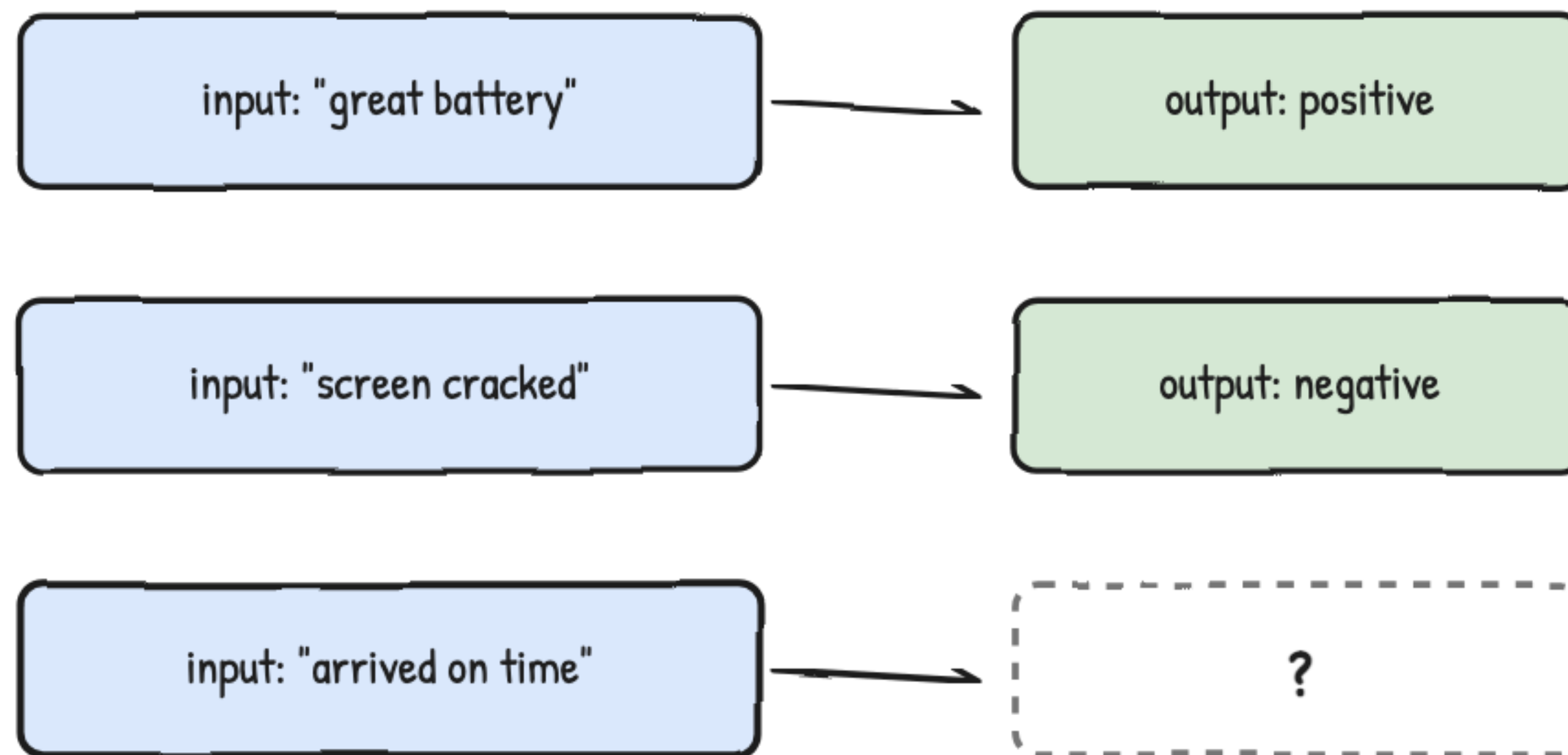
Prompt and context engineering, 2026 practice

A system prompt sets the standing rules for the whole conversation



The frame every other message sits inside

Few-shot examples show the pattern instead of describing it



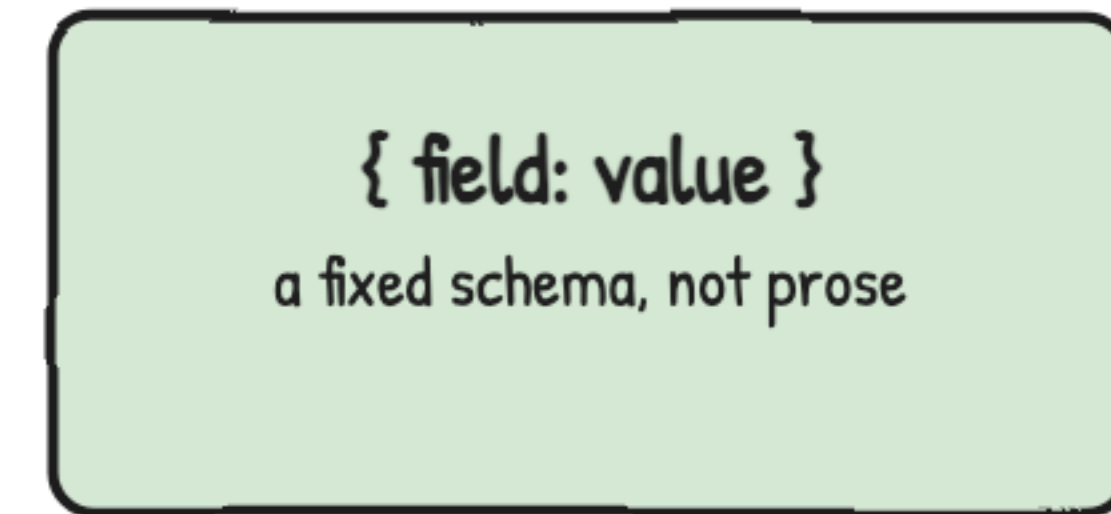
the model completes the pattern from just these examples

Two or three worked pairs, right inside the prompt

Structured output turns free text into something your code can parse



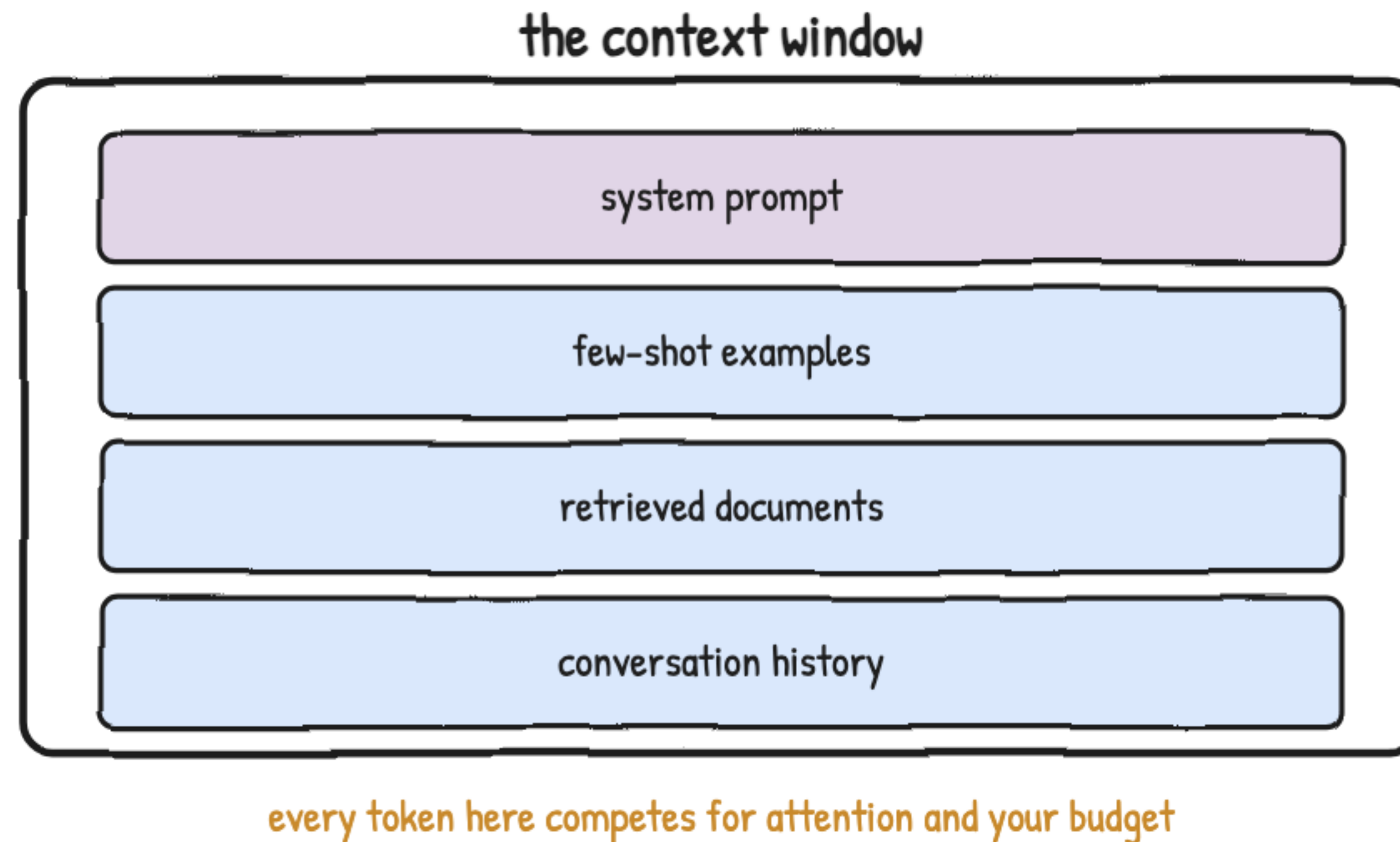
fragile regex to extract anything



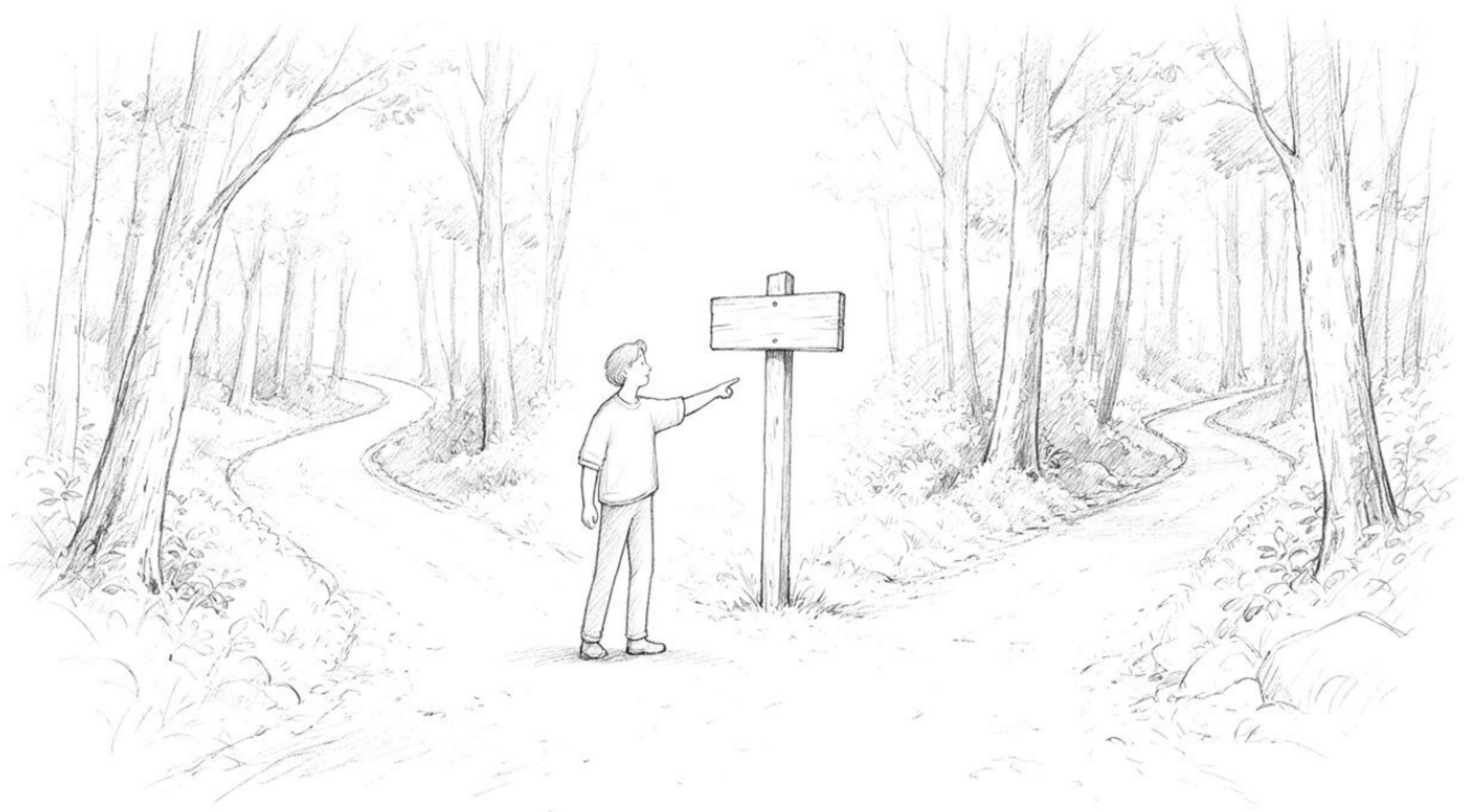
parses every time

Parses every time, instead of a fragile regex

Context engineering: everything in the window competes for attention and budget

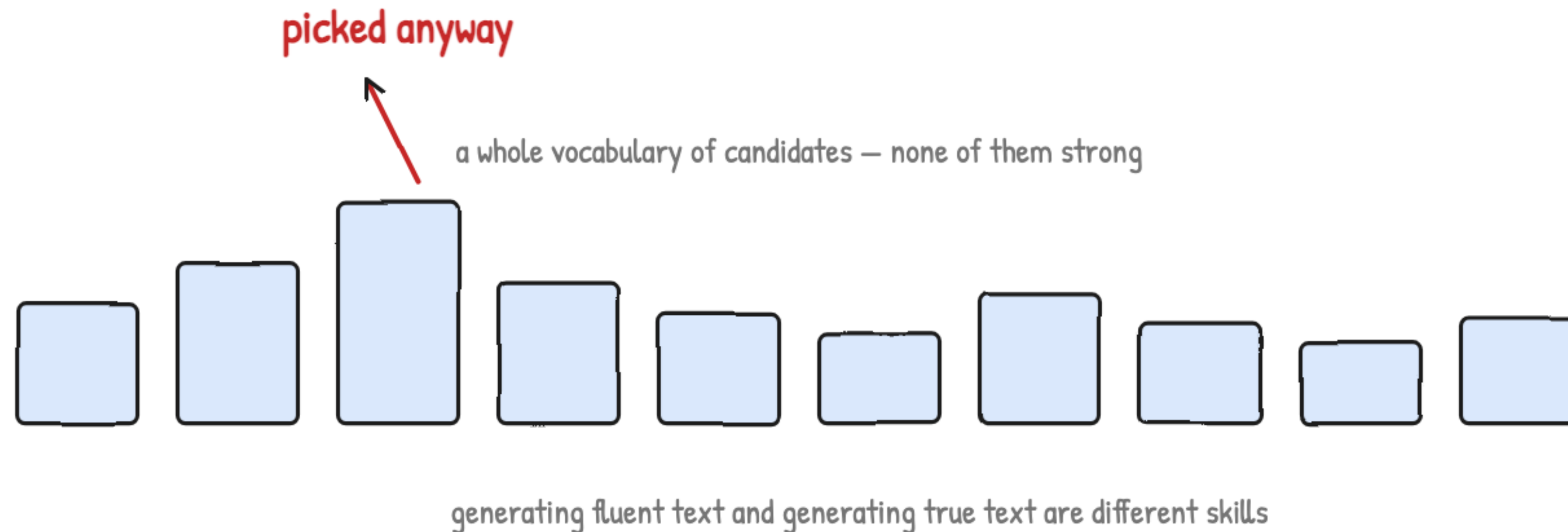


Deciding what goes in, not just what you write



Why LLMs hallucinate – the real reason

Every token forces a choice — there's no built-in "I don't know"



Fluent text and true text are different skills

The one question that catches most hallucinations

would the model have seen enough real
examples of this —
or is it the most fluent-sounding guess?

ask this before trusting a cited fact, a number, or a source

Ask this before trusting a cited fact, number, or source



RAG: chunk, embed, retrieve, stuff, generate

grep finds spelling; a colleague finds meaning



RAG gives a model that same colleague-level search

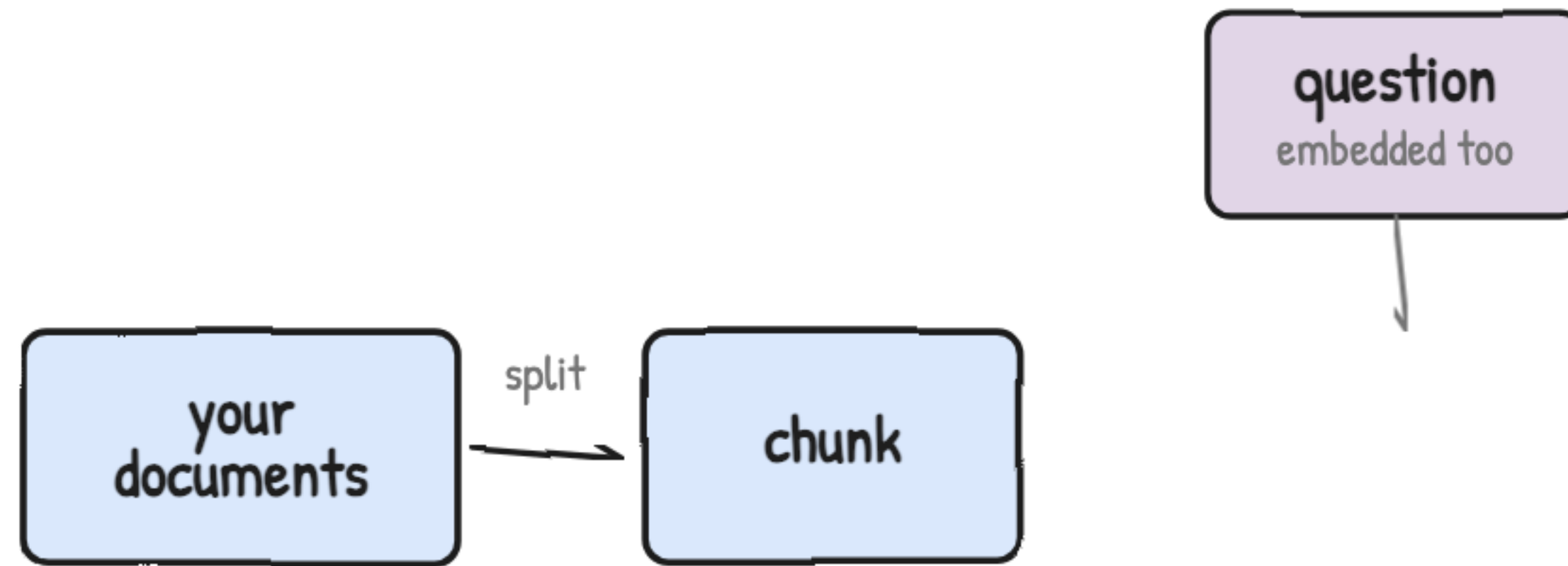
Chunk, embed, retrieve, stuff, generate — one pipeline, five hops



a weak answer is usually a retrieval problem, not a generation one

Your documents become a grounded answer

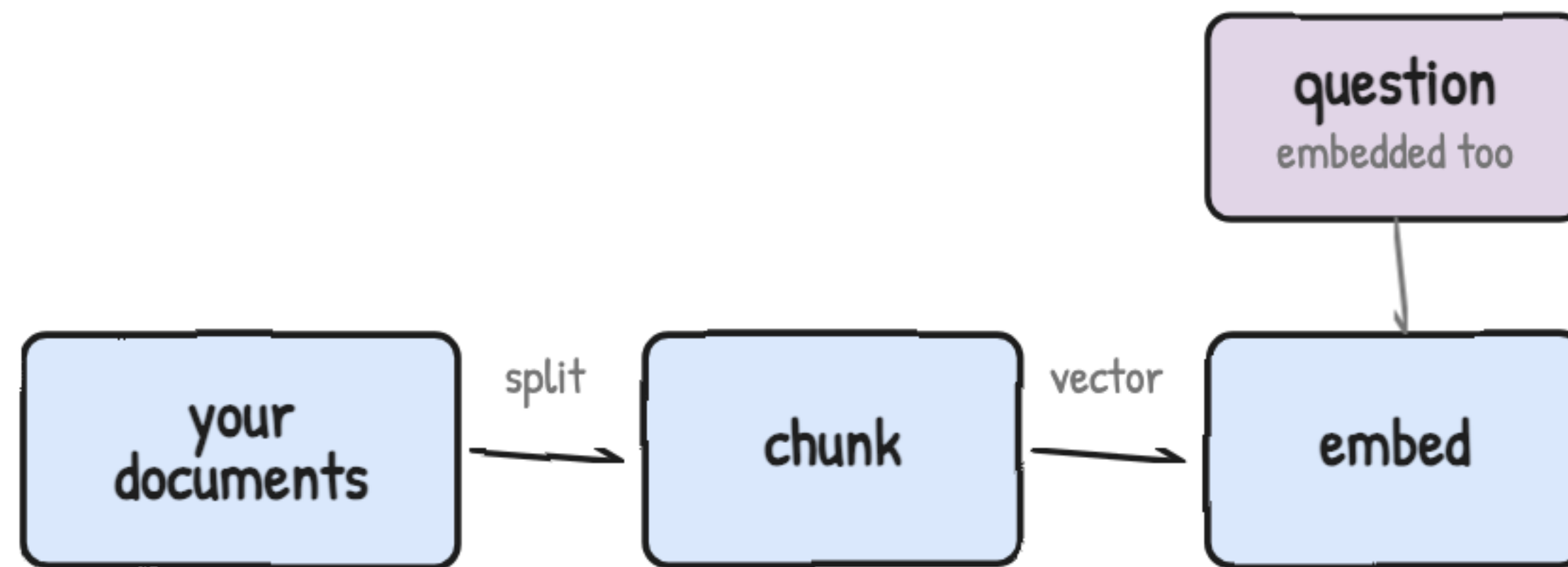
Chunk, embed, retrieve, stuff, generate — one pipeline, five hops



a weak answer is usually a retrieval problem, not a generation one

Your documents become a grounded answer

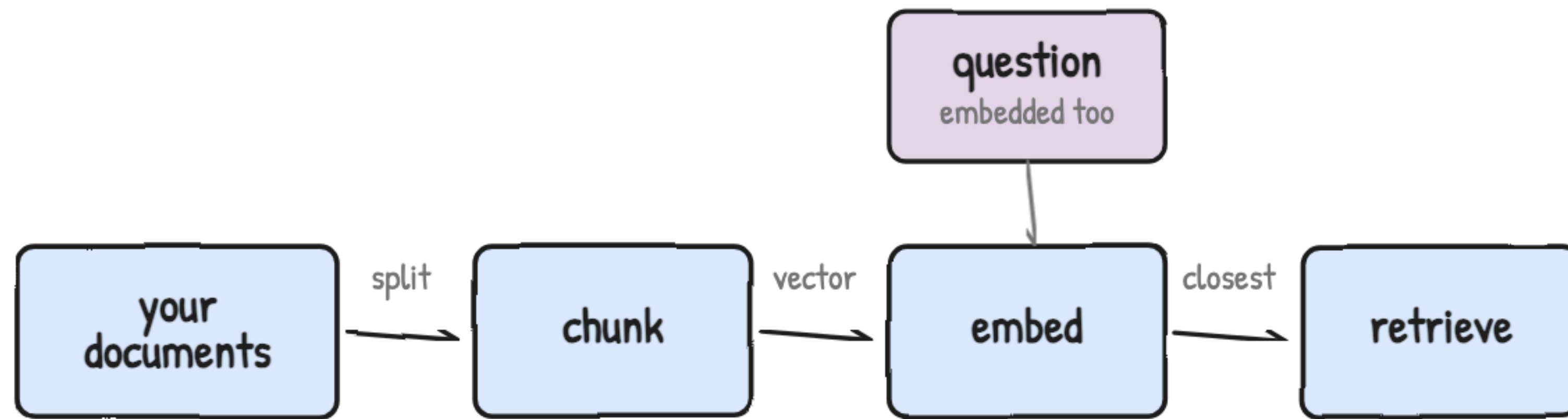
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Your documents become a grounded answer

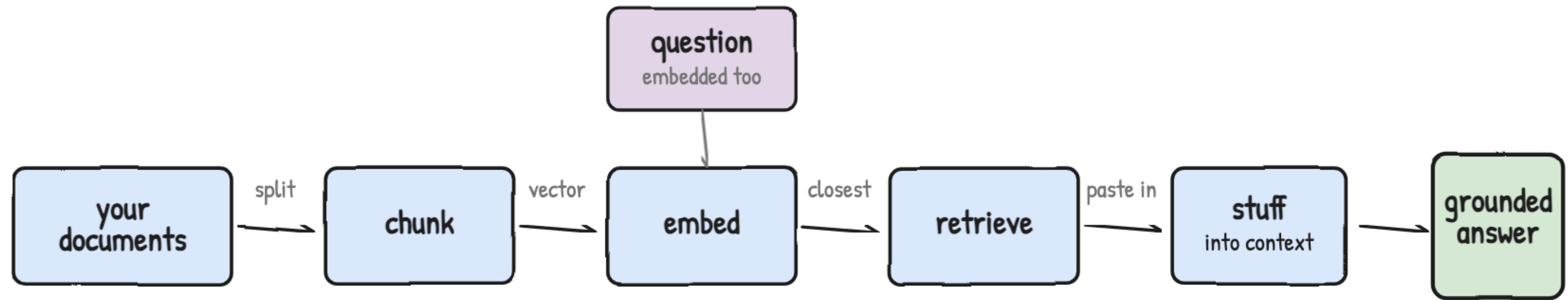
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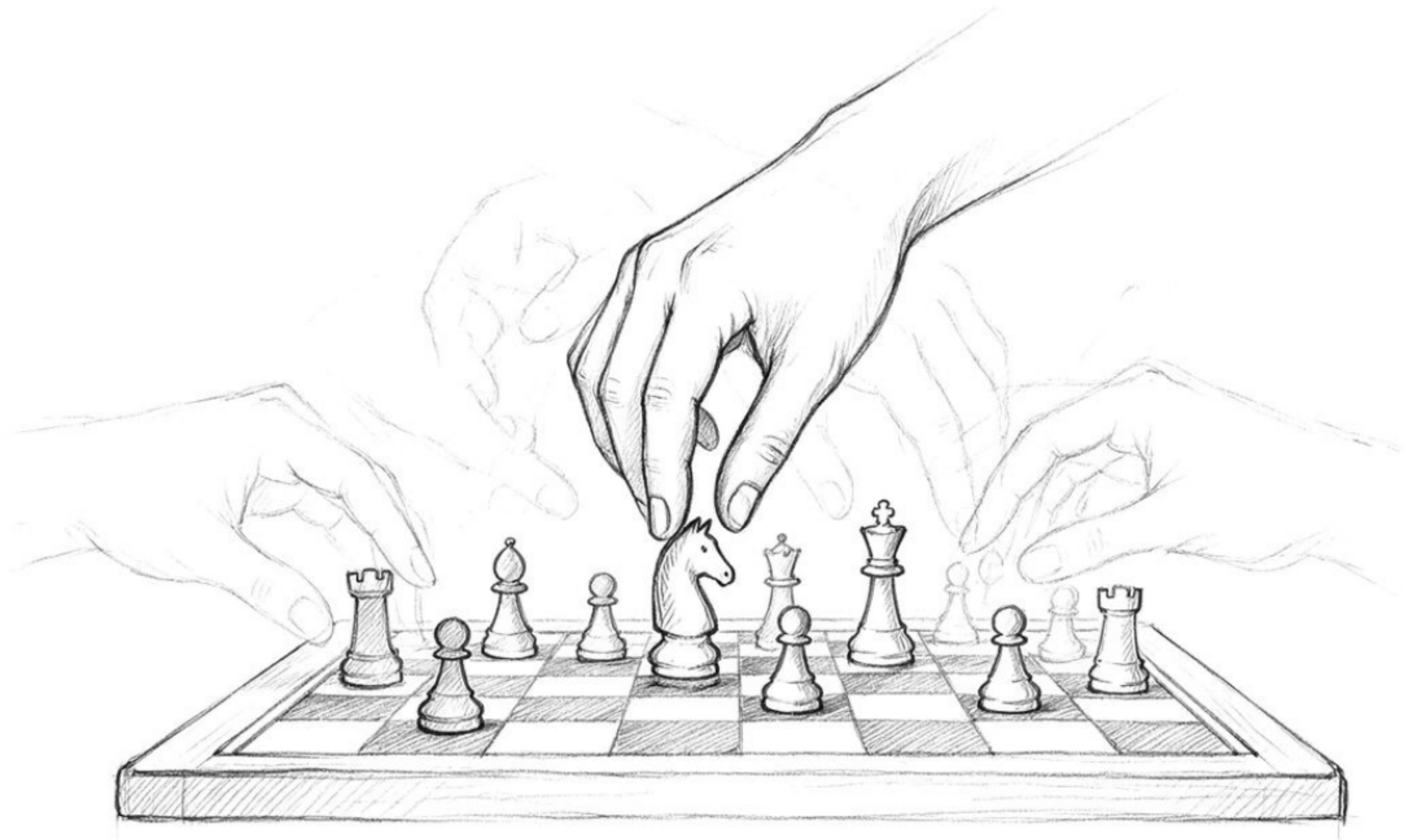
Your documents become a grounded answer

Chunk, embed, retrieve, stuff, generate — one pipeline, five hops



a weak answer is usually a retrieval problem, not a generation one

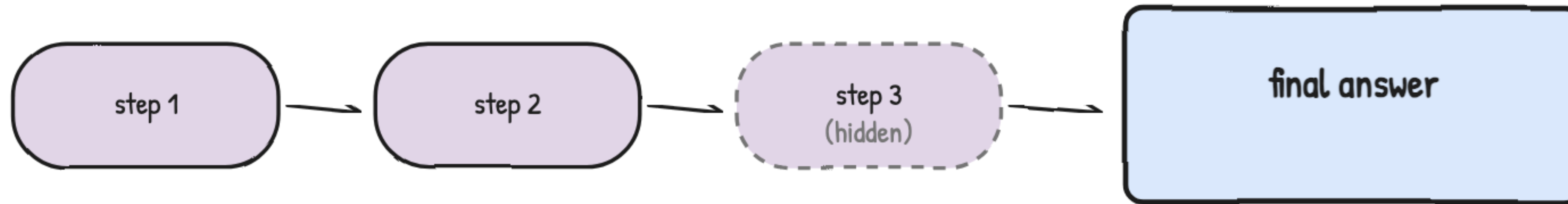
Your documents become a grounded answer



Reasoning models: what "thinking" costs you

A reasoning model spends tokens thinking before it answers

sometimes shown as a visible chain of steps, sometimes hidden and just billed



extra tokens, extra latency — every step

Sometimes visible, sometimes hidden and just billed

Turn it on for hard problems, off for simple lookups

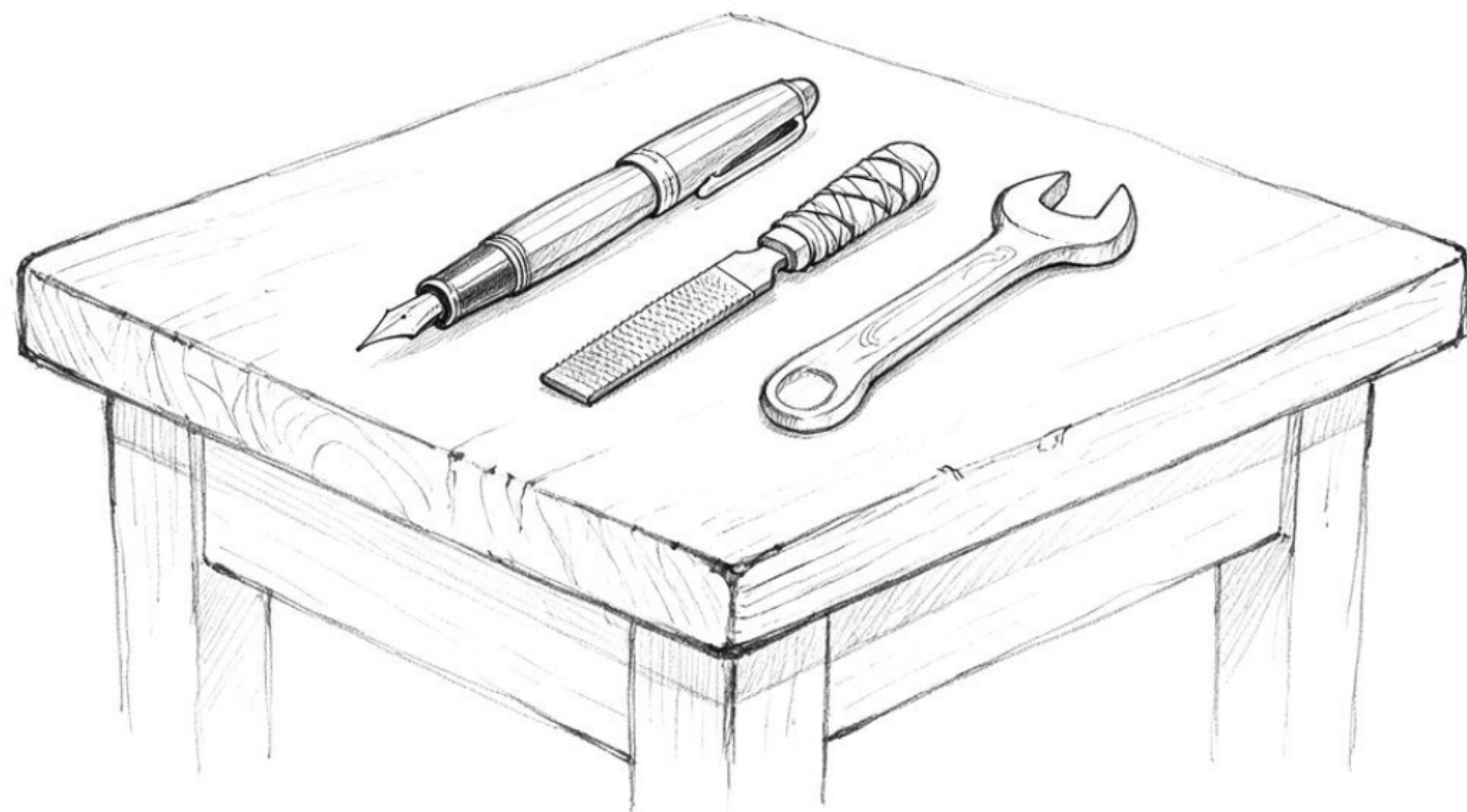
simple lookup
+ thinking mode ON

pure waste of money and latency

multi-step problem
+ thinking mode ON

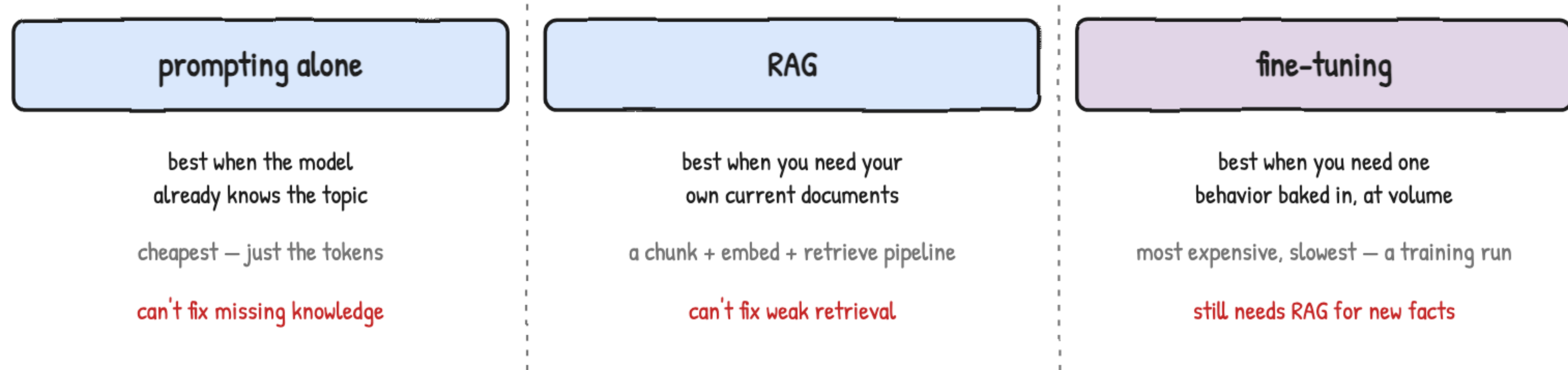
trades a wrong fast answer for a slower, reliable one

A cost-versus-quality decision, made per request



Decision guide: prompting vs RAG vs fine-tuning

Three approaches, three different jobs



not ranked best to worst — each answers a different question about what's missing

Not ranked best to worst — different jobs

MODULE 5 TAKEAWAYS

You now know what runs inside the model you call



- 1 tokens & parameters — the raw ingredients
- 2 attention & transformers — reads a sentence at once
- 3 pre-train, fine-tune, RLHF — how it got its shape
- 4 prompting, context, RAG — what you control and ground it in
- 5 reasoning & the decision guide — cost vs quality, per request

Now open Module 5 · Lab. · Gourav Shah · School of DevOps & AI